Scale without Scope: Firms Specialization and Industry Concentration

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Motivation

- Rising **industry concentration** in the U.S. over the past four decades
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  - *Product scope*: expand by entering new product markets
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- **This paper**:
  - Leading firms have become larger while **producing fewer products**
  - What drives leading firms to expand size while reducing product scope?
  - What does this pattern imply for **market power**, **productivity**, and **aggregate welfare**?

This Paper

1. Document new facts on **rising specialization** among industry leaders, 1990–2020.
   - Top 1% firms: average product scope declined by 40%.
   - Primarily due to continuing leaders shedding products

2. A general equilibrium model of multi-product firms under oligopolistic competition.
   - Firms choose the **number of products** & **production technology** for each product.
   - Scope and technology choices $\Leftrightarrow$ Market structure, productivity, and concentration

3. Estimation using U.S. firm-level data.
   - Macro trends in the U.S: $\uparrow$ concentration, $\downarrow$ **scope**
   - **Multi-product** & **technology choice** are important to understand the welfare effects

Relation to the Literature

- Macroeconomics of industry concentration & market power.
  De Loecker et al. (2020), Aghion et al. (2023), Edmond et al. (2023), De Ridder (2024), De Loecker et al. (2026)
  - Supply-side drivers of industry concentration: Haskel and Westlake (2018), Covarrubias et al. (2020), Autor et al. (2020), Crouzet et al. (2022), **Hsieh and Rossi-Hansberg (2023)**
  - Empirics: facts about long-term change in margins of firm expansion
  - Model: endogenous productivity choice & market structure
- Models of multi-product firms
  Klette & Kortum (2004), Eckel & Neary (2010), Bernard et al. (2010, 2011), Mayer et al. (2014), Nocke and Yeaple (2014), Mayer et al. (2021)
  - This paper: Highlight endogenous scope-size relationship by introducing intensive margin expansion through endogenous production technology choice.
- Literature on firm specialization
  Comment and Jarrell (1995), Stein (1997), Maksimovic and Phillips (2002), Ekerdt and Wu (2024), Ma (2026)

1. Descriptive Facts

Data

1. Dun & Bradstreet data 1990 - 2020: a near-universe of U.S. manufacturing firms
   - Products produced, employment, sales.

2. Definitions:
   - Industry: 2-digit SIC code.
   - Firm: A firm ID $\times$ industry. Results robust using alternative firm definitions.
   - Keep firms with $\geq$ 10 employees.
   - Product: A 8-digit SIC code, a total of $\sim$ 7,800 products in manufacturing sector.
     - Examples in electronic manufacturing: computer logic modules, solid-state memory, microprocessors
     - Examples in food and kindred products: packaged soft drinks, fresh fruit juices

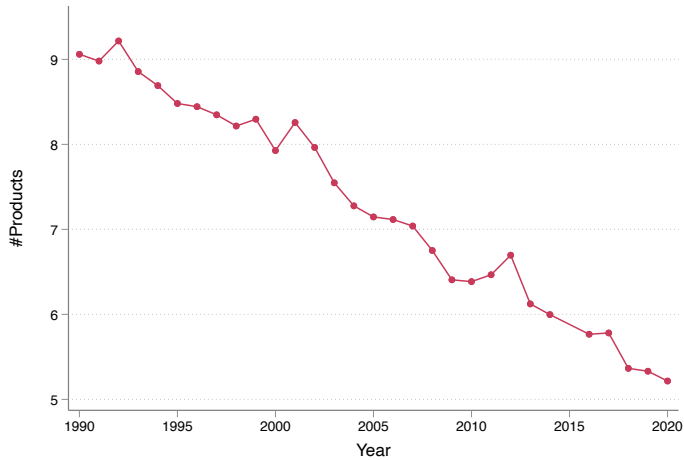
Products Intel Coca-Cola

Facts on Top Firm Specialization

1. **Industry leaders are increasingly specialized**
   - Average product scope of top 1% firms' declined by 40% — Facts
   - Mainly driven by incumbent leaders shedding product lines — Facts

2. Other related facts
   - Product overlap among top 1% firms goes down — Facts
   - Firms increasingly concentrate production to the best product — Facts

Product Scope of top 1% Firms Declines by 40%



Decomposing Changes in Product Scope

- **Employment-weighted average product scope of top 1% firms:**

$$N_t = \sum_{i \in \mathcal{T}_t} \omega_{it} n_{it}, \qquad \omega_{it} = \frac{\ell_{it}}{\sum_{j \in \mathcal{T}_t} \ell_{jt}}.$$

- **Dynamic decomposition between years t_1 and t_2** (Melitz & Polanec 2015):

$$\Delta N = \underbrace{\sum_{i \in C} \frac{\omega^C_{i,t_2} + \omega^C_{i,t_1}}{2} \Delta n_i}_{\text{Scope Adjustment}} + \underbrace{\sum_{i \in C} \frac{n_{i,t_2} + n_{i,t_1}}{2} \Delta \omega^C_i}_{\text{Employment Reallocation}} + \underbrace{\sum_{i \in E} \omega_{i,t_2} \left(n_{i,t_2} - N^C_{t_2}\right) - \sum_{i \in X} \omega_{i,t_1} \left(n_{i,t_1} - N^C_{t_1}\right)}_{\text{Net Entry / Exit}}$$

(1) Scope adjustment: continuing leaders actively shedding product lines ($\Delta n_i < 0$).

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(1) **Scope adjustment:** continuing leaders actively shedding product lines ($\Delta n_i < 0$).

(2) **Reallocation:** < 0 when emp share shifting toward narrower continuing leaders

(3) **Net entry/exit:** Churn in/out of top 1% (entrants E vs. exiters X relative to continuers N^C).

Scope Decline Mainly Driven by Continuing Leaders Shedding Products

Sample	Scope adjustment (1)	Emp reallocation (2)	Net entry (3)	Total change (4)
Top 1% firms	−2.36 (61.3%)	−0.98 (25.4%)	−0.51 (13.2%)	−3.84
Top 5% firms	−1.57 (53.7%)	−1.19 (40.6%)	−0.17 (5.7%)	−2.92
Top 10% firms	−1.37 (51.9%)	−1.21 (45.8%)	−0.06 (2.4%)	−2.64
Top 0.1% firms	−3.59 (66.3%)	−1.29 (23.9%)	−0.53 (9.8%)	−5.42
Top 20 firms	−2.72 (58.5%)	−1.23 (26.4%)	−0.70 (15.1%)	−4.65

- Within-firm scope adjustment plays an even larger role further up the firm-size distribution

Facts on Top Firm Specialization

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   - Mainly driven by incumbent leaders shedding product lines. Facts
2. Other related facts
   - Product overlap among top 1% firms goes down. Facts
   - Firms increasingly concentrate production to the best product Facts

- **Why are leading firms capturing larger industry shares while producing fewer products?**
- Next: A **supply-side** mechanism based on scalable technology investment

2. Model

Demand

$$\log U(Q) = \sum_{k=1}^{K} \alpha^k \log Q^k, \qquad \sum_{k=1}^{K} \alpha^k = 1$$

where $$Q^k = \left(\int_0^1 (q_j^k)^{\frac{\sigma-1}{\sigma}} dj \right)^{\frac{\sigma}{\sigma-1}}, \qquad q_j^k = \left(\sum_{i \in \mathcal{M}_j^k} (q_{ij}^k)^{\frac{\eta-1}{\eta}} \right)^{\frac{\eta}{\eta-1}}, \qquad \eta > \sigma > 1$$

- Households have Cobb-Douglas preferences over K industries, $k = \{1, 2, ..., K\}$.
- **Nested CES** structure within each industry (Atkeson & Burstein, 2008; Gaubert & Itskhoki, 2019).
  - Continuum of product markets $j \in [0, 1]$.
  - Horizontally differentiated varieties produced by a finite number of firms $\mathcal{M}_j^k$.

Production Technology

- Firm i produces product j using labor as the only input

$$q_{ij}^k = z_{ij}^k \ell_{ij}^k$$

where the productivity term is given by,

$$\log z_{ij}^k = \log A_i^k + \log a_{ij}^k + g\left(\psi x_{ij}^k\right)$$

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$$a_{ij}^k = \delta^{-n}$$

n is the **ranking** of product j (Eckel and Neary, 2010; Mayer et al., 2014)

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- Unit cost given by (wage normalized to 1): $c_{ij}^k = \frac{1}{z_{ij}^k} = \left(A_i^k a_{ij}^k \exp\left[g(\psi x_{ij}^k)\right]\right)^{-1}$

Timing

- Firms choose product scope, production technology, and pricing in two stages

- Stage 1: Each firm observes its firm-level productivity A_i^k, chooses product scope λ_i^k
  - Pay fixed product development cost F per product
  - Anonymous product market placement: randomly landed in λ_i^k product markets
  - Firm i understand its λ_i^k products will have fundamental productivity $\{A_i, A_i\delta^{-1}, \cdots A_i\delta^{1-\lambda_i^k}\}$

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- Stage 2: observe the realized market opportunities, choose investment and price for each product
  - Identity and fundamental productivity of all participants in each product market becomes public information

Stage 2 Problem: Price Competition and Product-level Investment

$$V(A_i^k, \omega_i^k) = \max_{\{p_{ij}^k, \ell_{ij}^k, x_{ij}^k\}_{j \in \mathcal{J}_i^k}} \sum_{j \in \mathcal{J}_i^k} \left[p_{ij}^k - c_{ij}^k(x_{ij}^k) \right] q_{ij}^k - C\left(\sum_{j \in \mathcal{J}_i^k} x_{ij}^k \right)$$

$$\text{s.t.} \qquad C\left(\sum_{j \in \mathcal{J}_i^k} x_{ij}^k \right) = \phi_1 \left(\sum_{j \in \mathcal{J}_i^k} x_{ij}^k \right)^{\gamma}, \quad \gamma \geq 1$$

- γ: curvature of investment cost
  - Parsimonious way of capturing increasingly costly firm expansion
  - $\gamma \to 1$: linear cost function. Firms make independent decisions across product markets
- ω_i^k: competition environment faced by firm i
  - Firm and product-level productivity of all rival producers linked to firm i

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- Within-market competition pins down the pricing rule

$$p_{ij}^k = \frac{\varepsilon_{ij}^k}{\varepsilon_{ij}^k - 1} \times c_{ij}^k = \frac{\varepsilon_{ij}^k}{\varepsilon_{ij}^k - 1} \times \frac{1}{z_{ij}^k}, \qquad \varepsilon_{ij}^k \equiv \varepsilon(s_{ij}^k) = \sigma s_{ij}^k + \eta(1 - s_{ij}^k).$$

  - Firms with higher productivity z_{ij}^k faces a lower elasticity of demand within a product market

Stage 1: Product Scope Choice

$$W\left(A_i^k\right) = \max_{\lambda_i^k \geq 0} \left\{ \mathbb{E}_{\omega \sim \Gamma^k(\lambda)} \left[V(A_i^k, \omega) \right] - F\lambda_i^k \right\}$$

where

$$\omega_i^k(n) \sim \Gamma^k(n), \quad \mathbf{\Gamma}^k \equiv \left\{ \Gamma^k(n) \right\}_{n=0}^{\lambda}.$$

- F is the product development cost.
- Each firm takes the distribution $\mathbf{\Gamma}^k = \{\Gamma^k(n)\}_{n \in \Lambda^k}$ as given
- Given the Perato distribution of firm-level productivity A_i^k, all firms make **symmetric actions**

$$\lambda_i^k(\cdot) \longleftrightarrow G^k = \text{Dist}(A, \lambda^k(A)) \xleftrightarrow{\text{random placement}} \mathbf{\Gamma}^k.$$

where

$$G^k(a, n) \equiv \Pr\left(A_i^k \leq a,\ \lambda_i^k \leq n\right) = \int_1^a 1\left\{\lambda^k(A) \leq n\right\} \vartheta A^{-\vartheta - 1}\, dA, \qquad a \geq 1.$$

Equilibrium Definition

A **Symmetric Bayesian Equilibrium** consists of

$$\{\lambda^k(A), G^k, \mathbf{\Gamma}^k\}_k, \qquad \{p_{ij}^k, x_{ij}^k, q_{ij}^k\}_{i,j,k}, \qquad \{P^k\}_k,\ P,\ Y,$$

satisfying:

1. *Optimality.* Given Γ^k, the scope policy solve Stage 1 problem. For realized environment, prices and investments solve Stage 2 problem.
2. *Consistency.* For every n, $\Gamma^k(n)$ is generated by $G^k = \text{Dist}\big(A, \lambda^k(A)\big)$.
3. *Market clearing.* Goods and labor markets clear.

$$\bar{L} = \sum_{k=1}^{K} N^k \int \left\{ F\lambda + \mathbb{E}_{\omega \sim \Gamma^k(\lambda)} \left[\sum_{j \in J_i^k} \ell_{ij}^k + C\left(X_i^k\right) \right] \right\} dG^k(A, \lambda).$$

3. Model Characterization

Model Mechanism: Optimal Investment & Product Scope

- **Optimal product-level investment (x_{ij}):**

$$\psi \times \underbrace{\ell_{ij}}_{\text{Total vairable cost}} \times g'(\psi x_{ij}) = \underbrace{C'(X_i)}_{\text{Marginal investment cost}} \quad ; \quad X_i \equiv \sum_{j \in \mathcal{J}_i} x_{ij}$$

  - **Scale-biased returns**: Upfront investment saves unit labor; returns scale with ℓ_{ij}.
  - **Internal cost linkage**: Total investment X_i sets the marginal cost of investment across all lines.

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- **Optimal product scope (λ_i):**

$$\pi_i(\lambda_i) = F + \underbrace{C'(X_i)\frac{\partial X_i}{\partial \lambda_i}}_{\text{additional investment expenditure}} + \underbrace{C'(X_i)\left(x_i(\lambda_i) - \frac{\partial X_i}{\partial \lambda_i}\right)}_{\text{profit forgone through investment crowd-out}}$$

  - An additional product crowds out investment in existing products

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Proposition

Suppose g is continuous at zero with $g(0) = 0$, twice continuously differentiable for $u > 0$, and satisfies $g'(u) > 0$ and $g''(u) + (\eta - 1)[g'(u)]^2 < 0$ for all $u > 0$. Firms with higher firm-level productivity A_i have greater total revenue and production employment, choose broader product scope, and make greater total investment.

Comparative Statics: A Rise in the Technology Efficiency ψ

$$\frac{d\lambda_i}{d\log\psi} = \frac{1}{(-\log\delta)}\left\{ \underbrace{\left[\psi x_i^{\lambda} g'(\psi x_i^{\lambda}) + \frac{d\log P_j}{d\log\psi}\right]}_{\text{within-product-market competition}} - \underbrace{\psi x_i^{\lambda} g'(\psi x_i^{\lambda}) \frac{d\log C'(X_i)}{d\log\psi}}_{\text{internal investment congestion}} \right\}$$

1. **Within-market competition:**
   - Productivity gains from $\uparrow \psi$ increases with initial investment
   - Tougher competition erodes market share in peripheral products and compresses operating profit

Details

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   - Tougher competition erodes market share in peripheral products and compresses operating profit
2. **Internal resource competition (within-firm crowding out)**
   - High A_i firms expand aggressively in core products $\Rightarrow \uparrow X_i$, crowding out peripheral products

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1. **Within-market competition:**

2. **Internal resource competition (within-firm crowding out)**

Proposition

Consider a small common increase in ψ, holding industry expenditure fixed. Suppose $\kappa(\psi x) = [g'(\psi x)]^2/[-g''(\psi x)]$ is strictly increasing,

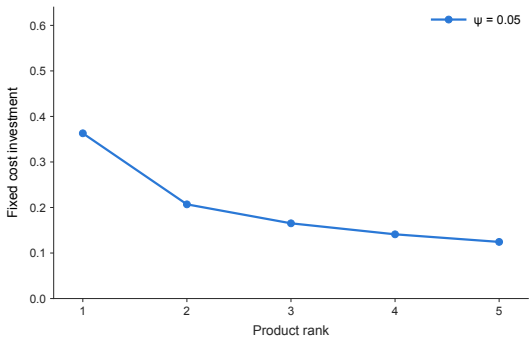
1. An increase in ψ generates larger productivity gains for suppliers with higher initial investment.
2. *Leading-firm specialization.* There exists a finite A^* such that firms with $A_i > A^*$ increase total investment and sales while reducing scope. Their marginal investment cost rises, and the industry sales share increases.

Details

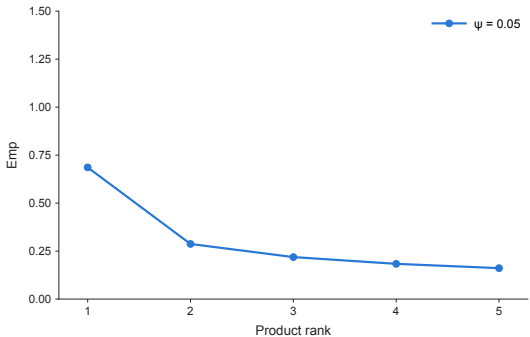
Numerical Example

Example:

$$g(\psi x_i) = \log(1 + x_{ij}^{\beta}), \quad 0 < \beta < 1$$



(a) Investment

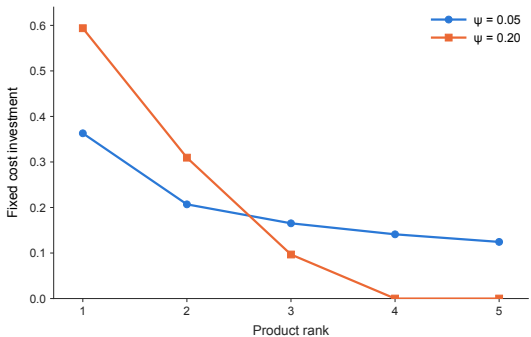


(b) Employment

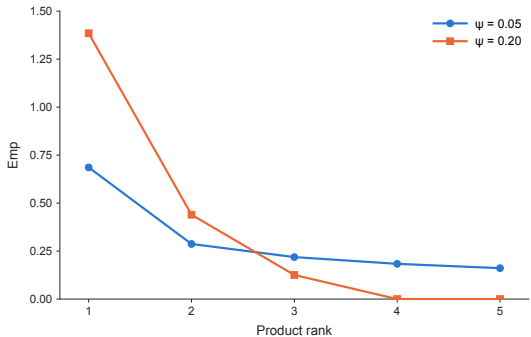
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(a) Investment



(b) Employment

Productivity Response to Changes in Technology Efficiency ψ

- Measured productivity in product market j is,

$$Z_j = \Big(\sum_{i \in \mathcal{M}_j} \Big(\frac{\mu_{ij}}{\mu_j} \Big)^{1-\eta} z_{ij}^{\eta-1} \Big)^{1/(\eta-1)}$$

- **Response to technology efficiency shock** is

$$\frac{d \log Z_j}{d \log \psi} = \underbrace{\sum_{i \in \mathcal{M}_j} \ell_{ij} \frac{d \log z_{ij}}{d \log \psi}}_{\text{technological improvement}} + \underbrace{\frac{\eta \bar{\zeta}_j}{\eta - 1} \text{Cov} \left(\frac{\mu_j}{\eta - (\eta - \sigma) s_{ij}}, \frac{d \log z_{ij}}{d \log \psi} \right)}_{\text{output reallocation}}. \tag{1}$$

  - $\bar{\zeta}_j$ captures the product-level average response of market share to relative productivity.
  - Production reallocation increases measured productivity if high markup firms invest more.

4. Quantification

Estimation Overview

- **Quantitative exercise**: estimate the model for 2020, study 1990 - 2020 changes

- Technology productivity function

$$g(x_{ij}) = \frac{1}{\eta - 1} \log\left(1 + [\psi x_{ij}]^{\beta}\right), \quad 0 < \beta < 1$$

  - ψ: normalized to 1 in the baseline calibration
  - Identify changes in ψ from 1990 to 2020 using changes in fixed-cost shares.

- Two groups of parameters
  - **Households** parameters: CES nests elasticities (Broda & Weinstein, 2006, Hottman et al.,2016)
  - **Production and investment cost** parameters (new) - SMM

Details

Estimation of the production and investment cost block

- Six parameters to estimate in the production and investment cost block:

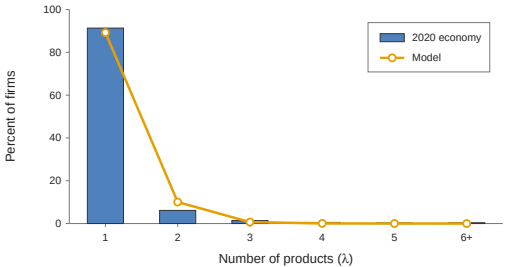
$$\Theta \equiv \{\vartheta, \delta, \beta, \phi_1, \gamma, F\}$$

- Two main variation used in the estimation
  - The **cross- and within-firm employment dispersion**
  - The fixed-cost share and variation by firm size

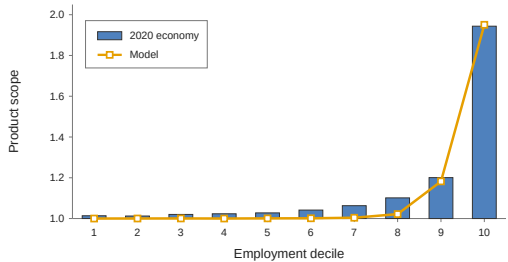
Moment	Value (Data/Model)
Firm-employment upper-tail slope	1.16 / 1.16
Share of multi-product firms	0.09 / 0.10
Core-to-second product emp gap: Bottom 50% firms	0.99 / 0.99
Core-to-second product emp gap: Top 10% - Bottom 50% firms	0.60 / 0.60
Revenue/variable cost in Compustat (R/VC)	1.55 / 1.48
Regression slope of log(R/VC) on log sales	0.03 / 0.03

Parameter values

Quantification: Model fit for untargetted moments



(a) Product scope



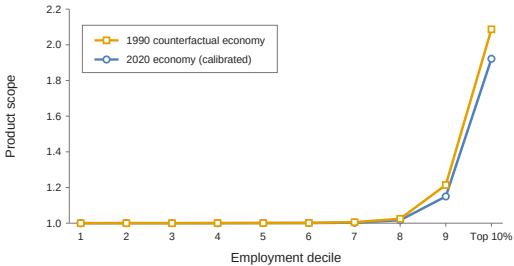
(b) Scope-employment relationship

- Firm-size distribution + scope-employment relationship is untargeted.
- Implied economy-wide markup: 1.32, within literature range ($1.1 \sim 1.4$)

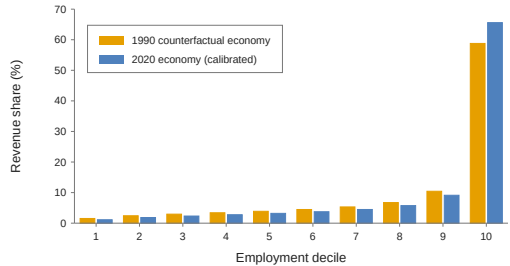
Changes in scope and size of industry leaders: 1990 - 2020

- Counterfactual: calibrate changes in ψ from 1990 to 2020 using changes in fixed-cost shares in data.
  - ψ drops by 38.61% from 2020 to 1990.

- The model captures well the change in industry leaders from 1990 to 2020 (1) reduction in product scope (2) higher industry share $\Longrightarrow$ **Scale up without being more broad**



(a) Product scope



(b) Revenue share

The impact of technological change on productivity and welfare

	Δ Model Counterfactual (%)	Δ Data (%)
A. Industry concentration		
Industry HHI	+21.53 %	+23.15%
Observed active firms	-4.91 %	-11.32 %
B. Product-market concentration		
Product market HHI	+30.29 %	–
Suppliers per product	-7.73 %	-11.55 %
D. Product scope		
Top-1% mean scope	-15.43 %	-36.04 %
Top-10% mean scope	-7.92 %	-16.15 %
E. Welfare and productivity		
Welfare, $Q = E/P$	+5.72 %	–
Productivity	+8.33 %	–

The role of firm adjustments and multi-product firms

- Decompose gains in aggregate welfare and productivity
  1. Direct gains from technology change: investment and scope fixed
  2. Single-product analogy: no changes in scope, uniform within-firm productivity growth
  3. Multi-product model with scope adjustment only
  4. Multi-product with scope adjustment and selective expansion

	Welfare	Productivity
CF 1. Direct gains from technology improvements	+3.95% (+3.95%)	+3.75% (+3.75%)
CF 2. Single-product analogy	+4.63% (+0.68%)	+5.13% (+1.38%)
CF 3. Multi-product with scope adjustment only	+5.01% (+0.38%)	+6.34% (+1.21%)
Full adjustment	+5.72% (+0.71%)	+8.33% (+1.99%)

- **Endogenous firm adjustments** generate sizable productivity (+4.58 pp) and welfare (+1.77 pp) gains
  - **Multi-product margin** accounts for $\sim 2/3$ of these gains.

Policy Counterfactuals

1. **Subsidize product development.** The government reimburses half of product development cost
   - Love of variety + stronger competition (Mankiw and Whinston 1986, Dhingra 2013)

2. **Correct product-level markup dispersion.** Tax or subsidize each supplier's revenue according to its within-market sales share s_{ij} (Edmond, Midrigan, and Xu 2023)

$$T_{ij}^{P} = p_{ij} q_{ij} \left\{ \frac{1}{\bar{\mu}_0} \left[1 + \frac{1 - (1 - s_{ij})^{(\sigma-1)/(\eta-1)}}{(\sigma - 1) s_{ij}} \right] - 1 \right\}.$$

   Every active supplier then charges $p_{ij}/c_{ij} = \bar{\mu}_0$, the baseline cost-weighted aggregate markup.

	Development subsidy (1)	Markup-targeting transfer (2)
Welfare	-1.05%	$+2.17\%$
Productivity	-0.21%	$+3.55\%$

Further evidence in support of model mechanism

- Focus on the relationship between information technology investment & firm product scope.
  - Fixed-cost investment in IT: digital engineering, ERP/SCM software, programmable automation.
  - IT as a share of corporate investment: 18% in 1980 to 35% in 2020.
    IT = Information-processing equipment + software

- **Prediction 1**: Industries experiencing faster improvements in IT efficiency ($\uparrow \psi^k$) should see a **larger decline in leading firms' product scope**.
  - Data: BEA/KLEMS industry ICT capital per worker (1997–2019) linked to D&B.

- **Prediction 2**: Individual firms that invest more heavily in IT **actively prune products**.
  - Data: Aberdeen CI Technology Database linked to D&B.
  - Firm-level measures of IT investment in hardware, software, and communication equipments.

Prediction 1 Prediction 2

Conclusion

- Empirically: Top 1% firms increase market share while producing 40% less products
  - Driven by incumbent leaders pruning products
- Theoretically: A model of multi-product firms with endogenous scope & technology choice
- Quantitatively: $\uparrow$ efficiency of scalable technology $\rightarrow$ Firm specialization + industry concentration
  - Important to model endogenous technology & scope choices
- Going forward: incorporating product churning + long-run growth

Thank you!

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Intel Product Dynamics in Electronic Equipment (SIC 36)

Product code	Short definition	1990	1995	2000	2005	2010	2015	2020
36740201	Computer logic modules	✓	✓	✓	✓	✓	✓	✓
36740204	Solid-state memory	✓	✓	✓	✓	✓	✓	✓
36740207	Microprocessors	✓	✓	✓	✓	✓	✓	✓
36740211	Semiconductor circuit networks	✓	✓	✓	✓	–	–	✓
36770000	Electronic coils/transformers	✓	–	–	–	–	–	–
36790102	Electronic commutators	✓	–	–	–	–	–	–
36720000	Printed circuit boards	✓	–	–	–	–	–	–
36740206	Integrated microcircuits	–	–	✓	✓	✓	✓	✓
36740200	Integrated circuits/networks	–	–	✓	✓	✓	✓	–
36610000	Telephone/telegraph equipment	–	–	✓	–	–	–	–
36619908	Fiber-optic communications equipment	–	–	–	✓	✓	✓	–
Product scope		7	4	7	7	6	6	5

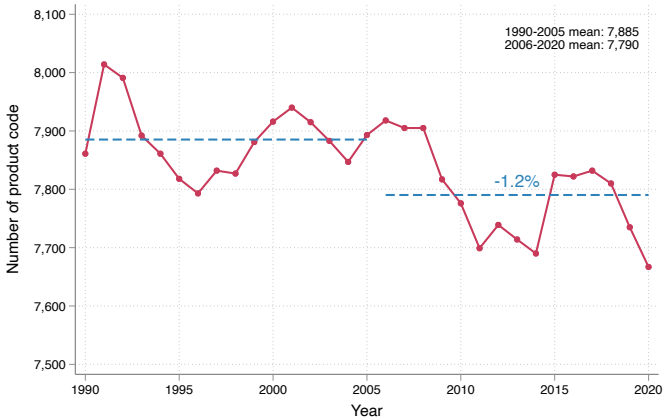
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Coca-Cola Product Dynamics in Food and Kindered Products (SIC 20)

Product code	Short definition	1990	1995	2000	2005	2010	2015	2020
20860301	Packaged nonalcoholic carbonated drinks	✓	✓	✓	✓	✓	✓	✓
20860302	Packaged soft drinks	✓	✓	✓	✓	✓	✓	✓
20870203	Drink concentrates	✓	✓	✓	✓	✓	✓	✓
20870208	Drink syrups	✓	✓	✓	✓	✓	✓	✓
20959904	Freeze-dried coffee	✓	✓	✓	✓	✓	✓	✓
20320102	Packaged baked beans with meat	✓	✓	–	–	–	–	–
20330304	Fresh fruit juices	–	–	✓	✓	–	–	–
20330305	Packaged fruit juices	✓	✓	✓	✓	✓	–	–
20370201	Frozen fruit-juice concentrates	✓	✓	✓	✓	✓	✓	–
20860101	Fruit drinks (less than 100% juice)	–	–	–	✓	–	✓	✓
20870201	Beverage bases	✓	✓	✓	✓	–	–	–
20879902	Flavoring concentrates (except drinks)	✓	–	✓	✓	✓	✓	✓
20959902	Coffee roasting	✓	–	–	–	–	–	–
Product scope		11	9	10	11	8	8	7

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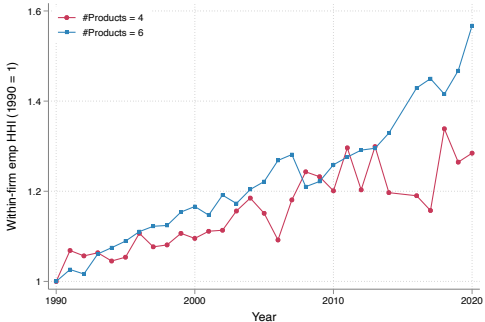
Number of Products by Year



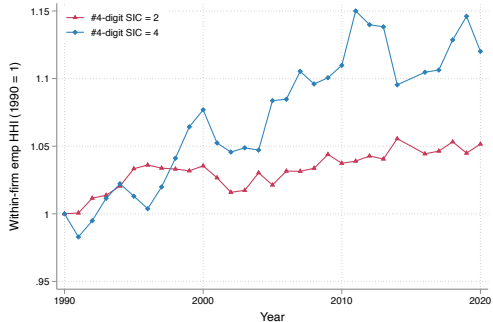
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Top 1% firms increasingly specialize in their best products

Compute within-firm employment HHI as: $HHI_i = \sum_{j \in \mathcal{J}_i} (\ell_{ij}/\ell_i)^2$



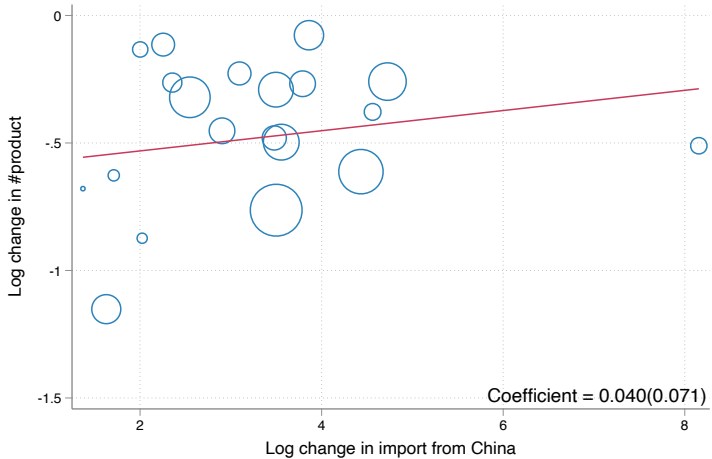
(a) HHI 8-digit SIC



(b) HHI 4-digit SIC

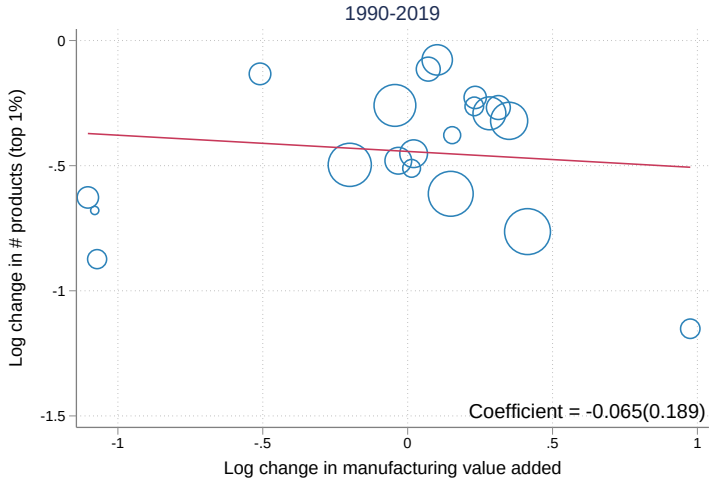
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China Shock is Not Strongly Correlated with Top 1% Scope Reduction



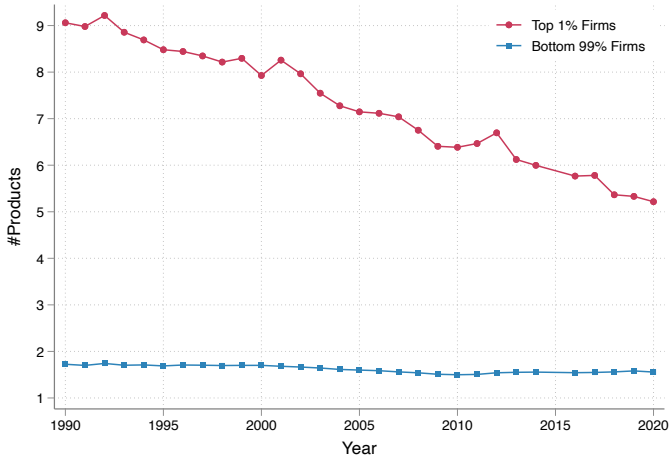
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Change in VA is Not Strongly Correlated with Top 1% Scope Reduction



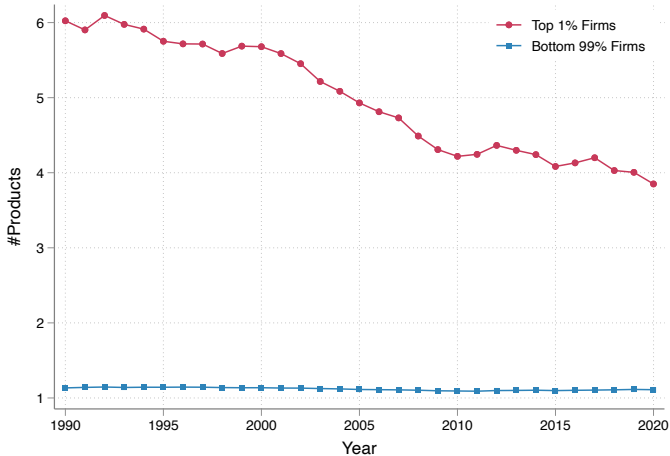
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Product scope among top 1% firms: Emp weight, Full



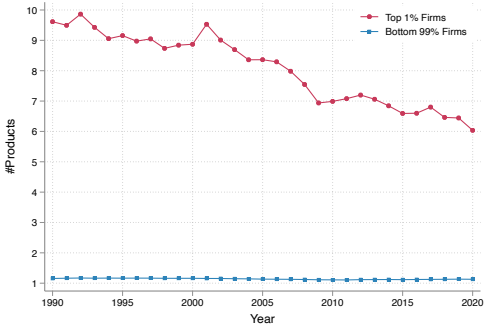
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Product scope among top 1% firms: No weight

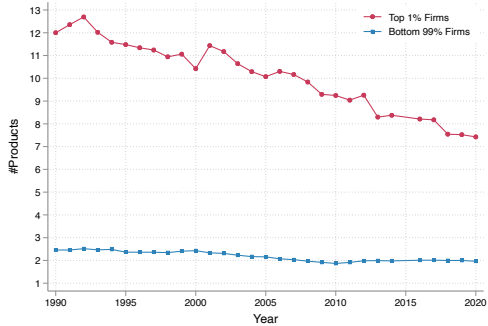


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Product scope among top 20 firms



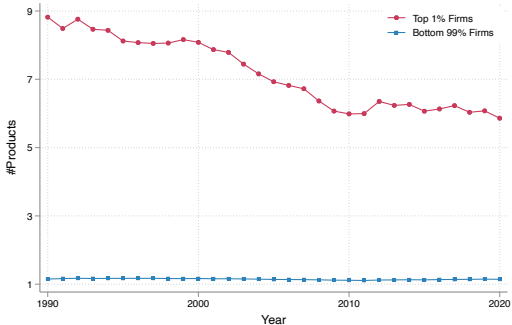
(a) Top 20 Firms: Unweighted



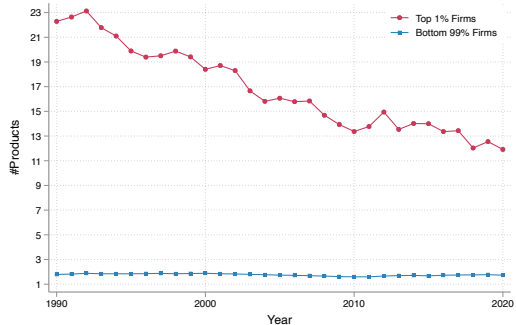
(b) Top 20 Firms: Emp weighted

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Product scope among top 1% firms (all industries)



(a) Top 1% Firms: Unweighted

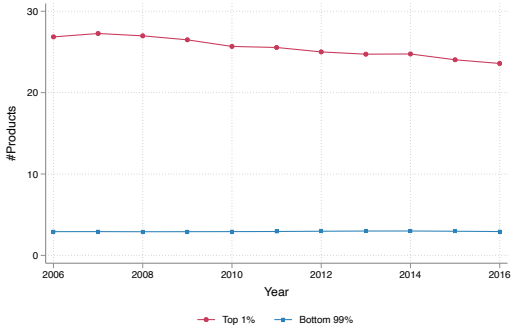


(b) Top 1% Firms: Emp weighted

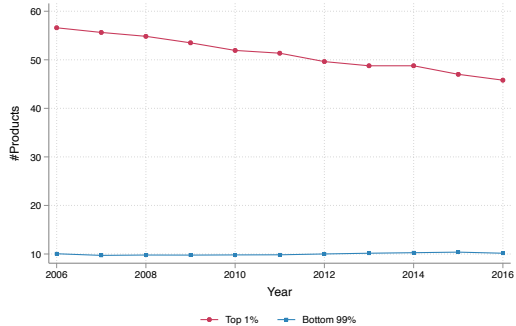
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Product Scope in the Nielsen Retail Scanner Data

Each product is a "product module" in Nielsen Retail Scanner dataset. E.g. "Cereal – Ready to Eat", "tooth cleaners".



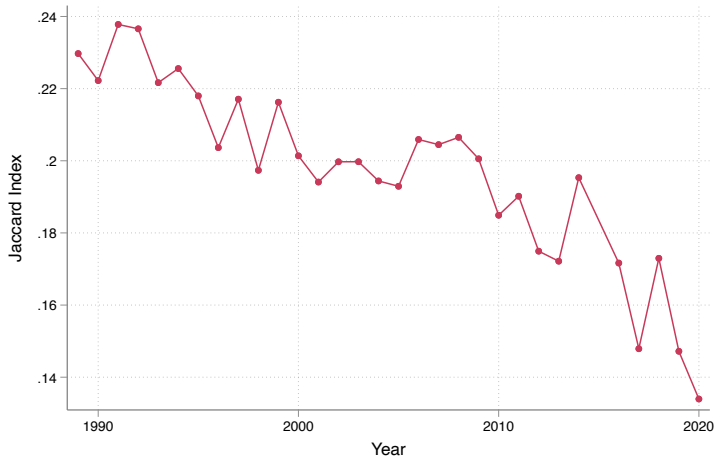
(a) Top 1% Firms: Unweighted



(b) Top 1% Firms: Sales weighted

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Product overlap of top 1% firms



$$J(S_i, S_{i'}) = \frac{|S_i \cap S_{i'}|}{|S_i \cup S_{i'}|}$$

- S_i: products in firm i's portfolio
- Pairwise overlap of product sets

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From scope choices to the firm distribution

Draw $A_i^k \sim \mathrm{Pareto}(1, \vartheta)$ and assign $\lambda_i^k = \lambda^k(A_i^k)$.

Joint distribution of productivity and product scope:

$$\boxed{\begin{aligned} G^k(a,n) &\equiv \Pr\Big(A_i^k \le a,\ \lambda_i^k \le n\Big) \\ &= \int_1^a 1\Big\{\lambda^k(A) \le n\Big\}\, \vartheta A^{-\vartheta-1}\, dA, \qquad a \ge 1. \end{aligned}}$$

Averaging over firms: for any integrable function f,

$$\int f(A,n)\, dG^k(A,n) = \int_1^\infty f\Big(A, \lambda^k(A)\Big)\, \vartheta A^{-\vartheta-1}\, dA.$$

G^k is induced by the scope policy. It is a probability distribution; N^k supplies the mass of firms.

From firms to competitive environments

$$\boxed{\Gamma^k(n) = \mathcal{R}_n\Big(G^k; N^k, \delta\Big), \qquad n \in \Lambda^k.}$$

$\mathcal{R}_n$ applies uniform random placement and records the *connected environment* of a focal firm with n products.

Number of rivals in a focal product market (large-market limit):

$$\#\big(\mathcal{M}_j^k \setminus \{i\}\big) \sim \text{Poisson}(\bar{n}^k), \qquad \bar{n}^k = N^k \int \lambda \, dG^k(A, \lambda).$$

Productivity and scope of an encountered rival:

$$d\widetilde{G}^k(A, \lambda) = \frac{\lambda}{\mathbb{E}_{G^k}[\lambda]} \, dG^k(A, \lambda).$$

Firms with more products are more likely to be encountered.

Conditional on scope λ, the rival's rank in the shared market is uniform on $\{0, \ldots, \lambda - 1\}$, with productivity $A\delta^d$. Follow its other products and their competitors to include indirect rivals.

Model Mechanism: Determinants of investment

Response of productivity

$$\frac{d \log z_{ij}}{d \log \psi} = \underbrace{\frac{\kappa(u_{ij})}{1-(\eta-1)\kappa(u_{ij})}}_{\text{technology responsiveness}} \underbrace{\left[1+(\eta-1)\frac{d \log P}{d \log \psi} - \frac{d \log \nu_i}{d \log \psi}\right]}_{\substack{\text{investment incentive net of} \\ \text{competition and internal congestion}}}.$$

Response of product scope

$$(-\log \delta)\frac{d\lambda_i}{d \log \psi} = \underbrace{\left[u_i^{\lambda} g'(u_i^{\lambda}) + \frac{d \log P}{d \log \psi}\right]}_{\substack{\text{within-product-market competition:} \\ \text{own gain relative to competitors}}} - \underbrace{u_i^{\lambda} g'(u_i^{\lambda}) \frac{d \log \nu_i}{d \log \psi}}_{\text{internal investment congestion}}.$$

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Model Mechanism: Value of Product Scope

- Value of additional products.

$$D_n \approx \pi_{j_{n+\mathbf{1}}} - F - \nu x_{j_{n+\mathbf{1}}}$$

- The change in the value of additional product can be written as

$$\frac{1}{\pi_{ij}} \frac{dD_{i,n}}{d\log\psi} \approx \underbrace{(\varepsilon_{ij}-1)u_{ij}g'(u_{ij})}_{\text{own technological benefit}} - \underbrace{\left[(\eta-\sigma)\sum_{\ell\in\mathcal{M}_j\setminus\{i\}} s_{\ell j}\left(G_{\ell j} - \frac{d\log\mu_{\ell j}}{d\log\psi}\right) + (\sigma-1)\overline{G}_P\right]}_{\text{external competition}} - \underbrace{(\gamma-1)\frac{\nu_i x_{ij}}{\pi_{ij}}\sum_{h\in\mathcal{J}_i}\omega^x_{ih}\frac{d\log x_{ih}}{d\log\psi}}_{\text{internal investment-cost channel}}. \tag{1}$$

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Household Parameters

Parameter	Interpretation	Source	Value
σ	EoS across product markets	Broda & Weinstein (2006)	2.5
η	EoS within product markets	Hottman, Redding, & Weinstein (2016)	5.0
N	Mass of potential entrants	Externally calibrated	30
ψ	Efficiency of production technology	Normalization	1.0

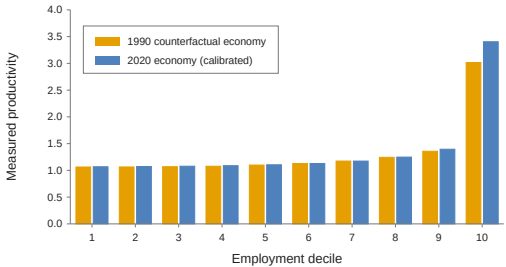
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Estimated Parameters

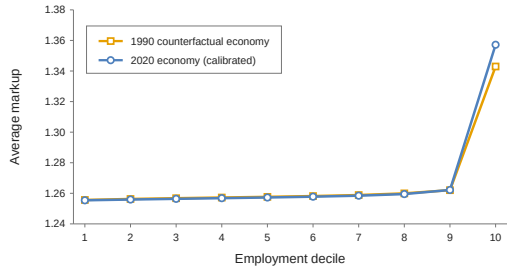
Parameter	Interpretation	Estimate
Firm productivity distribution and fixed cost		
ϑ	Pareto shape parameter	5.866
δ	Speed of productivity decay	0.717
F	Product development cost	0.013
Technology productivity and cost function		
β	Technology productivity curvature	0.902
ϕ_1	Investment cost scalar	0.011
γ	Investment cost curvature	1.161

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Changes in markup and productivity of industry leaders: 1990 - 2020



(a) Productivity



(b) Markup

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Prediction 1: Industry IT Exposure and Top-Firm Scope

- Proxy industry-level IT efficiency shock using changes in IT per worker

$$\Delta \text{Exposure}_{k,t} = \log\left(\frac{\text{ICT Capital}_{k,t}}{L_{k,t}}\right) - \log\left(\frac{\text{ICT Capital}_{k,t-1}}{L_{k,t-1}}\right),$$

  - *Assumption:* Sectors with higher potential to deploy IT undergo faster capital deepening when IT efficiency improves.

- Empirical specification:

$$\Delta y_{k,t} = \alpha \Delta \text{Exposure}_{k,t} + X'_{k,t}\beta + \gamma_k + \varepsilon_{k,t}.$$

  - $\Delta y_{k,t}$: Log change in product scope among **top 1% firms** in industry k between year t and $t-1$.
  - Industry FE γ_k.
  - $X_{k,t}$ controls for industry concentration, value-added, import competition

Prediction 1: Firms in Industries with Greater IT Adoption Specialize More

	Outcome: Δ Scope of top 1% firms		
	(1)	(2)	(3)
Δ Log IT per worker	-0.357***	-0.290**	-0.310***
	(0.137)	(0.108)	(0.103)
Initial log HHI		0.091***	0.094***
		(0.020)	(0.019)
Δ HHI		0.109***	0.099***
		(0.037)	(0.034)
Δ log(VA)		0.241	0.136
		(0.167)	(0.179)
Δ log(China Import)			0.010
			(0.077)
Observations	460	460	460
R-squared	0.071	0.101	0.103

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Prediction 2: IT Investment and Firm Product Scope

- Relate real annual IT investment to subsequent changes in product scope:

$$\Delta y_{ikt} = \alpha \log I_{ik,t} + X'_{ik,t}\beta + \mu_{ik} + \varepsilon_{ik,t+1}.$$

  - Δy_{ikt}: (1) Change in product scope; (2) Log change in employment of firm i in industry k between year t and $t-1$.
  - $I_{ik,t}$: real annual IT investment between year t and $t-1$.
  - $X_{ik,t}$: control for firm size in year $t-1$ (log capital, employment, number of products).
  - Firm, age, & industry $\times$ year fixed effects.

Prediction 2: IT Spending and Firm Product Scope

	Δ Scope		Δ log Employment	
	(1) Full	(2) Top 1%	(3) Full	(4) Top 1%
Log real IT investment	-0.006*	-0.014**	0.014	0.037*
	(0.003)	(0.006)	(0.010)	(0.019)
Log employment, $t-1$	0.087***	0.093***	-0.481***	-0.515***
	(0.013)	(0.019)	(0.028)	(0.045)
Log real IT capital, $t-1$	-0.006	-0.006	-0.023*	-0.014
	(0.004)	(0.006)	(0.013)	(0.020)
Firm FE	Yes	Yes	Yes	Yes
SIC2 × year FE	Yes	Yes	Yes	Yes
Age FE	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes
Observations	911,762	18,353	911,762	18,353

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